AI651

Deep Learning for Space, Time and Graphs

Assignment 02

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Github

Question 1: River Forecasting Proposal

Architecture Diagram

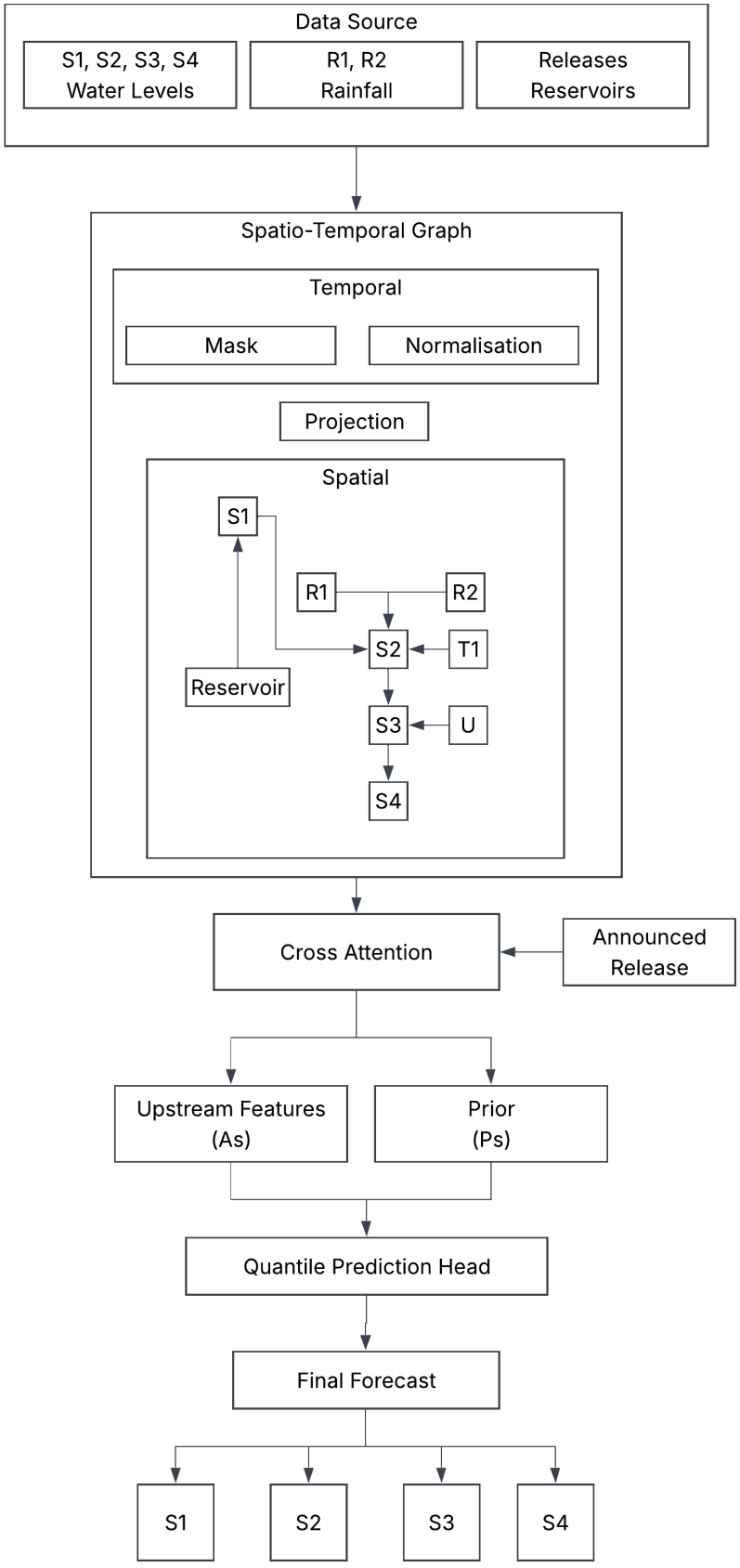


Table 1 Tensors

Tensor	Shape
S1, S2, S3, S4, T1 (each)	[B, L, 5, 4]
R1, R2 (each)	[B, L, 2, 4]
Reservoir	[B, L, 1, 5]
Forecast	[B, 4, 3]

Where

- B is the batch size
- L is the lookback window (later defined as 120)
- The last column defines the features

Assumptions

1. All data sources are coordinated in time; they have the same resolution, the internal clocks align, and they are synchronized.
2. For the reservoirs, the release time is dependable; the announced release is followed through to avoid missing measurements (later identified through a mask)
3. Normalization is not done on a station level to avoid inconsistency.
4. All stations are sampled at the same rate
5. Time delay is accommodated using the delay function defined below.
6. No one station overpowers the rest; the data collected is valued equally for each station.
7. Downtime for any data source is no longer than 10 minutes to avoid multiple missed measurements.
8. Downstream water level is affected by environmental conditions; rainwater may cause a change in water level that is not sourced from the neighboring sensor.
9. A latent node used to represent the unmonitored tributary; the inflow is non-negative
10. A late measurement is considered as a missing sample and does not affect future records; it is not assumed to be the current value at t+10.
11. The direction of water flow starts from S1 → S4 and does not flow back to a previous station.
12. Distance between stations is not identical so the travel time varies as well.

Forecasting and Data Specifications

Issuance time starts at t_o recorded in minutes and a forecast is issued after every 10 minutes for $t_o + 60$. Since we do not have a defined window, we can assume a retention time of 120 steps to accommodate any missed measurements and also to start detecting patterns. The actual number can be learned as a parameter but for the sake of this formulation, 120 step window which is 1200 minutes since each step is sampled after every 10 minutes.

The historical window is $h_w = [t_o - 1200]$ and quartile definition is based on pinball loss to be $q = \{0.1, 0.5, 0.9\}$

One hour horizon is defined by $Y^{pred}(t_o, t_o + 60)$

Table 2 Forecasting Data Specification

	Shape	*Axes Description*	*Timestamps/Coverage*
Station	[B, 120, 5, 4]	Batch, time, stations, features	t_o , $t_o + 60$
Rainfall	[B, 120, 2, 4]	Batch, time, gauge, features	random
Historical release	[B, 120, 1, 5]	Batch, time, reservoir, features	t_o - 1200 , t_o)
Reservoir	[B, 6, 5]	Batch, horizon, features	t_o
Forecast	[B, 4, 3]	Batch, stations, quantiles	$t_o + 60$

Information available at issuance

- Water level measurements
- Rainfall measurements
- Reservoir release schedule

Model computations and design justification

First we define the final forecast equation

$$\hat{y}_{s,q}(t_0 + 60) = x_s^{\text{last}} + P_s + \sigma_s g_{s,(q)}$$

- x^{last} is the latest available observed water level at station s.
- P_s is the upstream propagation
- σ_s is a positive station-specific uncertainty scale.
- $g_{s,(q)}$ is the ordered quantile adjustment

Next we define the training objective

$$\mathcal{L} = \frac{1}{B} \sum_b \sum_s \sum_q \rho_q \left(\frac{y_{b,s} - \hat{y}_{b,s,q}}{\sigma_s} \right)$$

Where

- $y_{b,s}$ is the actual water level
- $y^{hat}_{b,s}$ is the predicted water level
- σ_s is a normalization factor

And quantile loss

$$\rho_q(u) = \max(qu, (q-1)u)$$

For the Spatio-Temporal Graph first we define the temporal

$$T_{\tau,i}^{(\ell)} = \sum_{k=0}^{2} W_k^{(\ell)} S_{\tau-k\delta_\ell,i}^{(\ell)}.$$

Then the spatial component

$$S_{\tau,i}^{(\ell)} = \sum_r \sum_{j \in \mathcal{N}_r(i)} g_{r,ij} \odot Z_{\tau,j}^{(\ell-1)}.$$

Then combining them to form the spatio-temporal backbone

$$Z_{\tau,i}^{(\ell)} = Z_{\tau,i}^{(\ell-1)} + \mathrm{SiLU}\left(\mathrm{LN}\left(T_{\tau,i}^{(\ell)}\right)\right).$$

With the delay defined as

$$\delta_\ell = 2^{\ell-1}.$$

Operation and exceptional conditions

i. Missing values are treated by a mask that defines that the zero value is not the actual water level but that the measurement is unavailable. If value exists, then mask = 1 otherwise mask = 0. Multiplication operation applies the mask.

ii. Delayed observations are not incorporated into the next iteration as the current value, rather they are used in the historical context of the next timestamp

iii. The announced future release must be timed before t_0 and the time of release is used for the prediction of the hour that it falls into.

iv. The discrepancy between scheduled and actual release time is penalized beyond the sample rate (10 minutes). Any delay lesser than this is ignore and if it is exceeded, it is used in the previous or subsequent iteration.

v. The proposed model makes use of as little historical context as possible in order to lower the computation and storage costs. Running every 10 minutes requires only 6 steps to be used into the hourly prediction thereby keeping the compute costs low.

Uncertainty, limitations, and additional information

The main source of uncertainty is the environmental conditions of the stationed area that affected by natural factors which cannot be accurately defined or deduced from the information available; the climate changes and external factors can lead to unexpected changes. Another source is the unmonitored tributary from where the inflows that are not recorded will have an impact on stations S3 and S4 which may be hard to expect simply by using other stations; this architecture proposes a latent variable to lower the uncertainty level.

Additional sources of information that can help the model are a rain gauge on the unmonitored tributary, and adding more stations beyond the affected area to provide better insight into the river's water levels.

Question 2: Motion or Memory

Initial check before run

Table 3

CHECK	EXPECTED	MINE
(A) MAX|M(M(X)) – X|	0	0.0 (equal = True)
(B) MAX|R^4(X) – X|	0	0.0 (equal = True)
(C) MAX|MRM(X) – R^{-1}(X)|	0	0.0 (equal = True)
(D) BRIGHT PIXEL OF M(P)	(7, 10, 43)	(7, 10, 43)
(E) BRIGHT PIXEL OF R(P)	(7, 43, 10)	(7, 43, 10)
(F) 1× INPUT / TARGET FRAMES	1–5 / 6–10	[1, 2, 3, 4, 5] / [6, 7, 8, 9, 10]
(G) 2× INPUT / TARGET FRAMES	1, 3, 5, 7, 9 / 11, 13, 15, 17, 19	[1, 3, 5, 7, 9] / [11, 13, 15, 17, 19]
(H) 3× INPUT / TARGET FRAMES	1, 4, 7, 10, 13 / 16, 19	[1, 4, 7, 10, 13] / [16, 19]

Q 2.1

Based on figure 1a, **mirrored** transformations are not familiar to the model; across the values of theta, the pixels (y-axis) are not following a repeated distribution showing that a mirror transformation on a bin is interpreted as an entirely new bin of data. for example, bin 0-20 (theta) with a value of ~0.049 mirrors to bin 180-200 which has a value of ~0.08 when both the values should have been equal or with minimal difference for it to be perfectly adapted to mirrored data. Similarly for **rotation**, the histogram does not prove a complete adaptation to rotation with a 90 degrees angle; bin 120-140 have a value of ~0.03 and when rotated by 90, bin 210-220 has a value near 0.07. Neither transformation is fully familiar to the model after training on this data. The training data cannot be proven to be symmetric at this point.

From **figure 1b**, a typical speed would reasonably be based on the median value that is 3.32. At 2x, and 3x, the values of speed are ~6.6 with pixels <0.01 and ~9.9 with no recorded values for pixels.

Since overlapping digits violate the optical flow's Spatial Smoothness assumption that neighboring pixels tend to belong to the same surface and move similarly, the histogram will obtain a bias that changes the analysis about whether the data is already symmetric or not.

Figure 1a: histogram of flow direction

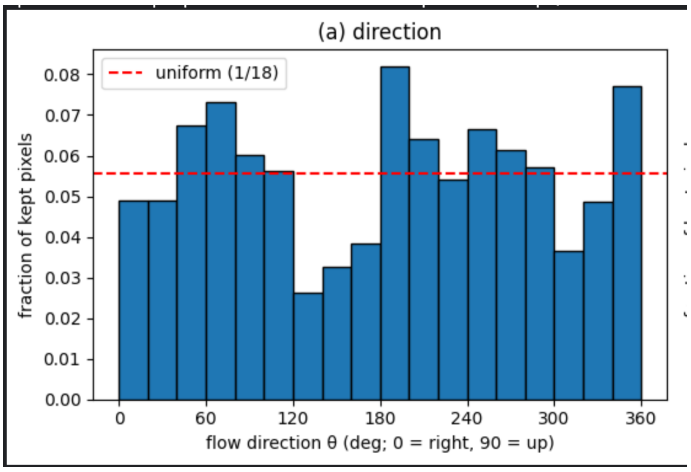


Figure 1 a: histogram of flow direction

Figure 1b: histogram of flow speed

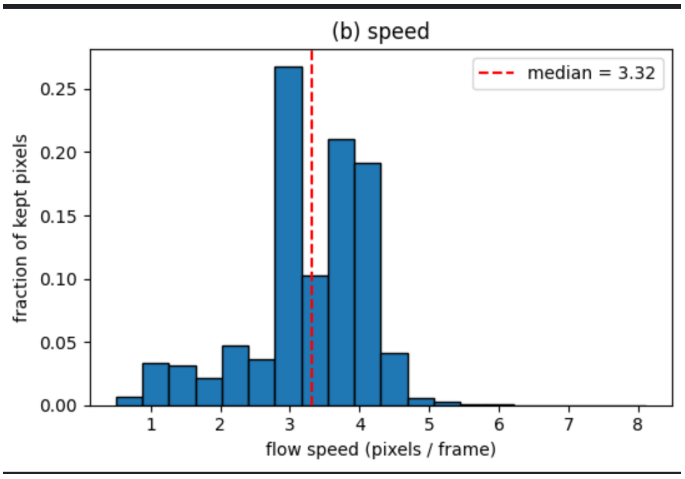


Figure 1b: histogram of flow speed

Q 2.2

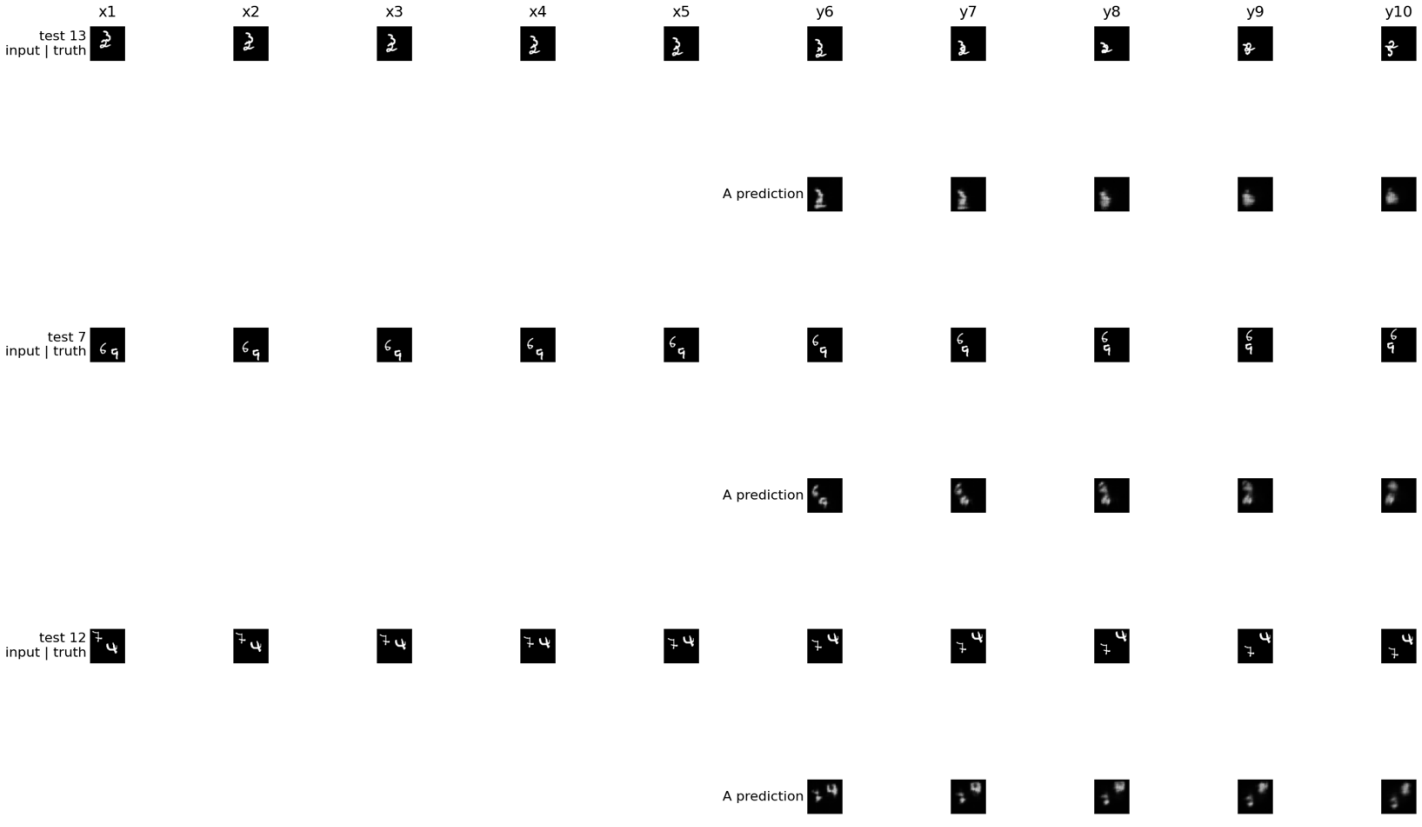
Rotation and mirror change both the motion statistics and digit appearance while speed-up only changes the motion statistics. Since two of them test the appearance at different levels, we can test whether the level of change impacts the ability of the model and by making the third one only affect the motion, we can test on a separate aspect of the transformation's impact. By combining all 3 in different ways, we deduce the impact of each type of transformation on the model.

Baseline ConvLSTM

Table 2

MODEL	MSE1	MSE2	MSE3	MSE4	MSE5	E(5)	PARAMS
BLANK	0.0426	0.0426	0.0426	0.0425	0.0425	0.0426	-
COPY-LAST	0.0492	0.0604	0.0669	0.0708	0.0732	0.0641	-
A	0.0167	0.0200	0.0223	0.0241	0.0257	0.0218	480,784

Figure 2 : test videos



Q 2.3

Reach: roughly 4 pixels per time step considering the 4 step stride of 16x16 filter.

Q 2.4

A's MSE is significantly lower then both Blank and Copy-Last predictors across all horizon values with the largest difference at horizon value = 5. The error grows from 0.167 to 0.257 over the horizons. When digits overlap, model A's performance also suffers as we observe around y8 in the predictions.

2.5

Considering A's performance and reach, I would predict for it to struggle on both mirror and rotation since those two transformations significantly change how the image looks compared to what the model has learned during training. For speed however, I think it will at least pass until 2x when normal speed is at 3.32 though it might fail at 3x since it also uses convolution which does not carry much information forward across time beyond the window

Testing the Baseline

Table 3

MODEL	E(5)	C_M	E_M	C_R	E_R
BLANK	0.0426	0	0.0426	0	0.0426
COPY-LAST	0.0641	0	0.0641	0	0.0641
A	0.0218 (0.0227, 0.0209)	0.00215 (0.00187, 0.00244)	0.0225 (0.0233, 0.0217)	0.00359 (0.00365, 0.00353)	0.0237 (0.0245, 0.0229)

Figure 3

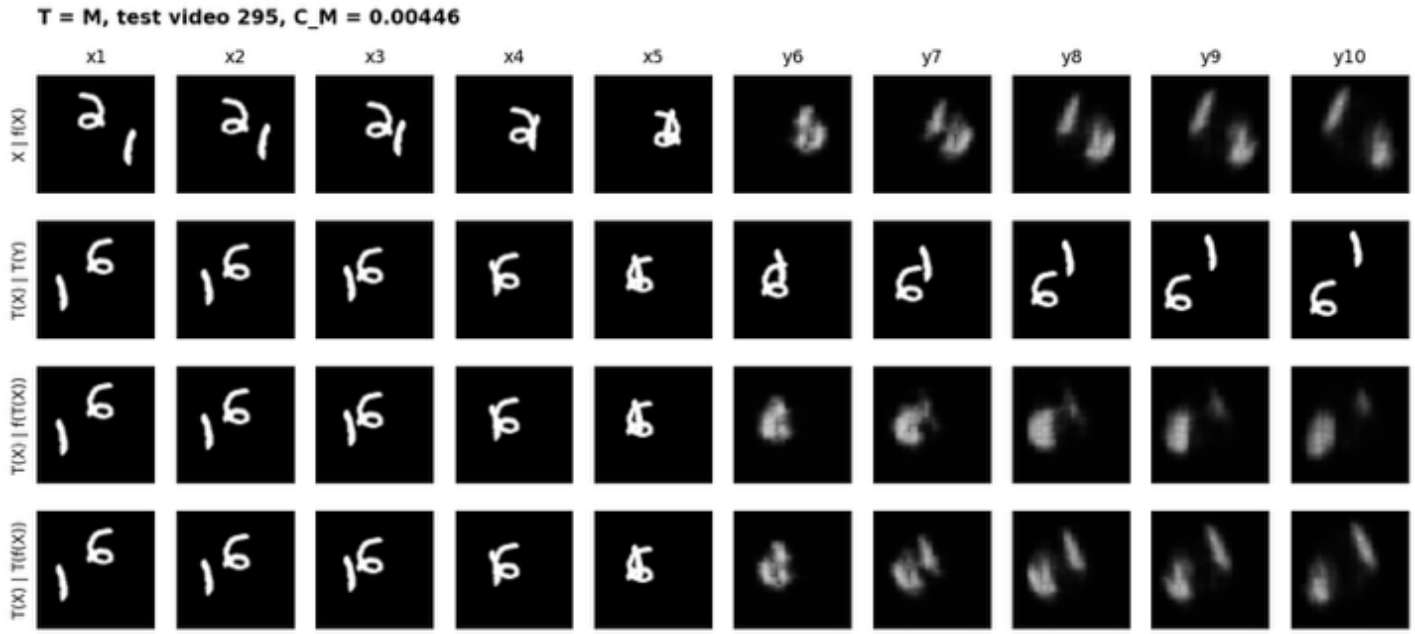
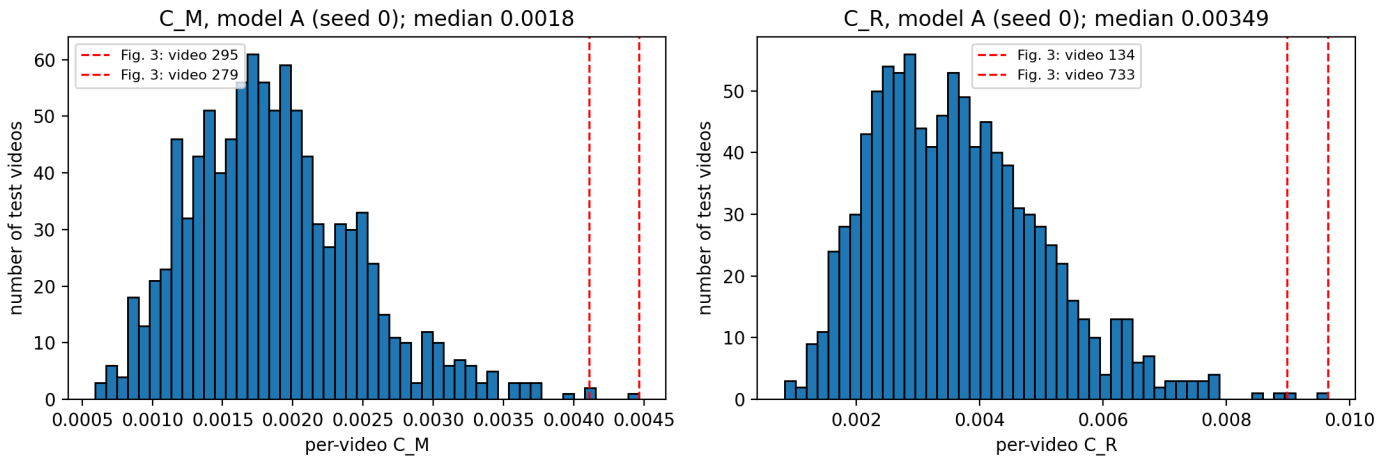


Figure 4



2.6

Consistency can be misleading since it seems like the model is learning well but in reality it is showing that model's performance does not deviate much; from previous analysis, we know that mirror and rotated forecasts are laden with errors and looking at a small Ct with this context knowledge shows that the model is consistently performing poorly. This is why Et is crucial to quantify the error and find out whether the model is actually learning or not.

2.7

In Figure 3, row 3 and 4 are inconstent in terms of the digit's actual shape. Row 3 blurs our earlier than row 4 which maintains the two separate objects until later.

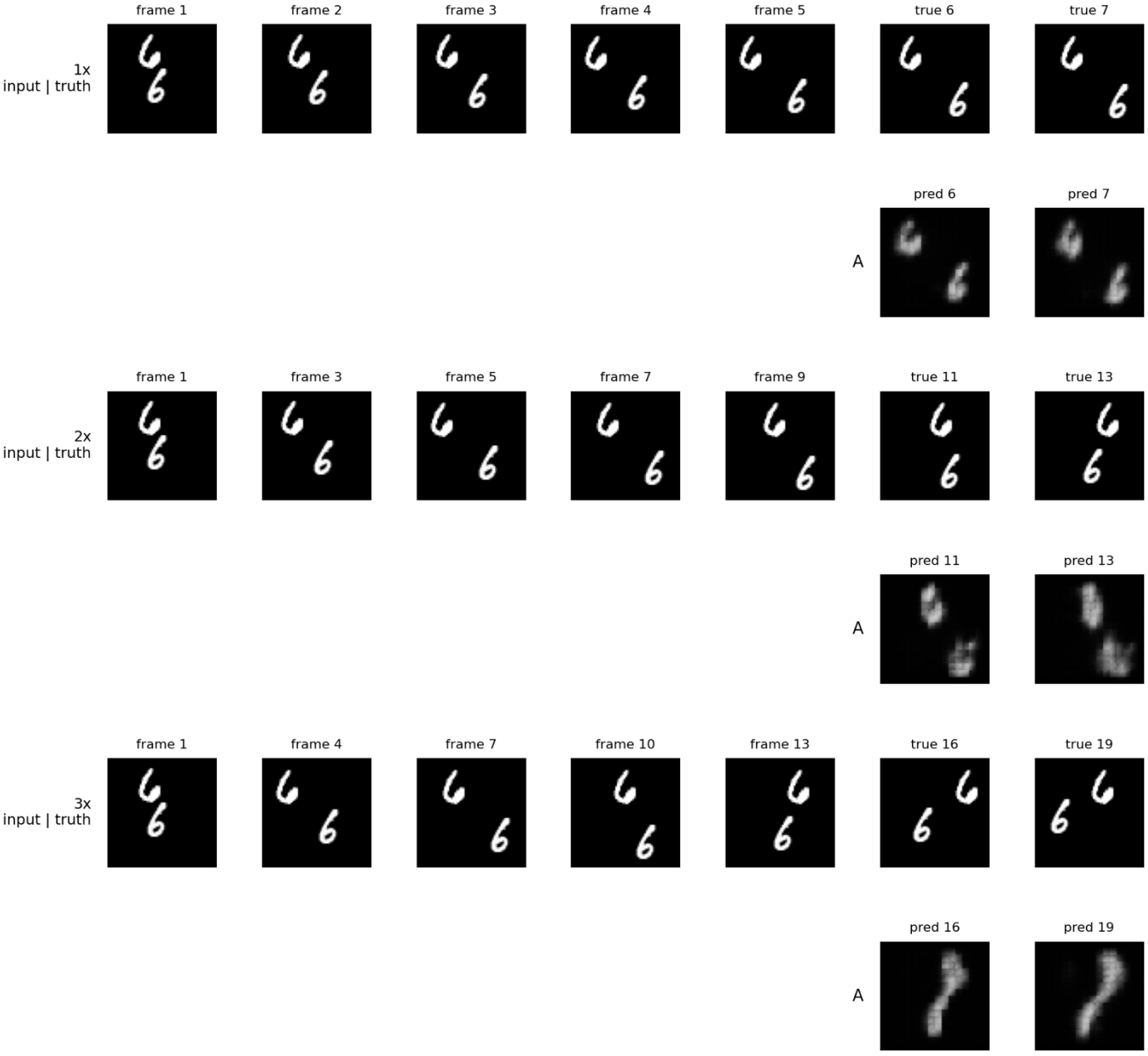
2.8

According to figure 3, rotation seems to produce a slightly higher error rate than mirroring. Since the main changes in rotation are related to the appearance more than mirror as stated in 2.2, we can argue that the error rises from the unprecedented appearance while the error from motion remains rather menial.

Table 4

MODEL	E(2) 1X	E(2) 2X	E(2) 3X	E(5) 1X	E(5) 2X
COPY-LAST	0.0548	0.0661	0.0703	0.0641	0.0719
BLANK	0.0426	0.0425	0.0426	0.0426	0.0425
A	0.0184 (0.0192, 0.0175)	0.0447 (0.0447, 0.0447)	0.0517 (0.0531, 0.0504)	0.0218 (0.0227, 0.0209)	0.0499 (0.0509, 0.0489)

Figure 5



2.9

The model's predictions start to fail as early as 2x which does not align with the previous predictions but the failure at 3x is proven right although after failing at 3x, it is to be expected. The failed predicitons at 2x are blurry but they are able to capture the location and gross size of the objects using the previous frames. At 3x, the digit entirley vanishes and the model predicts rather like the blank predictor.

2.10

Similar to the technique with transformations, the easiest way to decide whether speed up caused the increase in failure rate or reach contributed more, is to train with both conditions being explored separately. This

means that there must be data that is sped up and also a newer version of the model that has a longer reach. Then, running a comparison test between the two will help in the identification of each factor's contribution.

What Can Data Fix?

2.11

B: fixes appearance-based failures and mirror and rotation failure

C: fixes failure due to sped up data

D: fixes failure due to smaller reach along with speed uncertainty

Table 5

MODEL	E(5) 1X	C_M	E_M	C_R	E_R	E(2) 1X	E(2) 2X	E(2) 3X	PARAMS	TRAINS/EPOCH	INFERS
A	0.0218	0.00215	0.0225	0.00359	0.0237	0.0184	0.0447	0.0517	480,784	10.7	0.423
B	0.0229	0.00126	0.0229	0.00178	0.0229	0.0191	0.0446	0.0519	480,784	11.2	0.422
C	0.0276	0.00144	0.0278	0.00262	0.0287	0.0235	0.0291	0.0335	480,784	10.6	0.422
D	0.0293	8.47e-04	0.0295	0.00165	0.0302	0.0270	0.0297	0.0320	480,784	18.6	0.712

2.12

Reach increases to 8 pixels which is double the value of model A. we start with 3x3 filters, with a patch size of 8 and 64 dim leading to a reach of 8.

2.13

Appearance: symmetry augmentation in B

Unfamiliar motion: 2x and 3x speed in C and D which worsens performance compared to reference predictor Blank.

Architectural reach: D not performing that better than C.

C is not failing at 3x while D performs well which shows that pushing more data without improving the model's architecture does not guarantee an improved result. The main issue occurred due to the limited window across a much faster speed which is likely to be resolved by increasing the receptive field rather than the amount of data.

2.14

The changes did cost accuracy with the best trade for B by significantly reducing Cm and Cr to almost half the original value. C follows and conceptually adds more value as it deals with varying speeds which we have

observed to worsen the performance previously. A and D do cost accuracy but they are arguably not worth the improved stats especially since D requires a lot more training resources for little gain.

The Final Model

Table 6

MODEL	E(5) 1X	C_M (10^-17)	E_M	C_R (10^-17)	E_R	E(2) 1X	E(2) 2X	E(2) 3X	PARAMS	TRAIN S/EPOCH	INFERS
SYM-B	0.022	9.00	0.022	7.29	0.022	0.018	0.043	0.050	480,784	11.2	3.56
SYM-D	0.029	5.30	0.029	4.02	0.029	0.027	0.029	0.031	480,784	18.6	5.82
SYM-DB	0.028	6.29	0.028	4.95	0.028	0.025	0.028	0.031	480,784	19.1	5.83

Based on table 6, final model is should be sym-B for shorter horizons and sym-DB for longer horizons. Since we are also trying to work around the speedup problem, D+B with symmetry is a better choice.

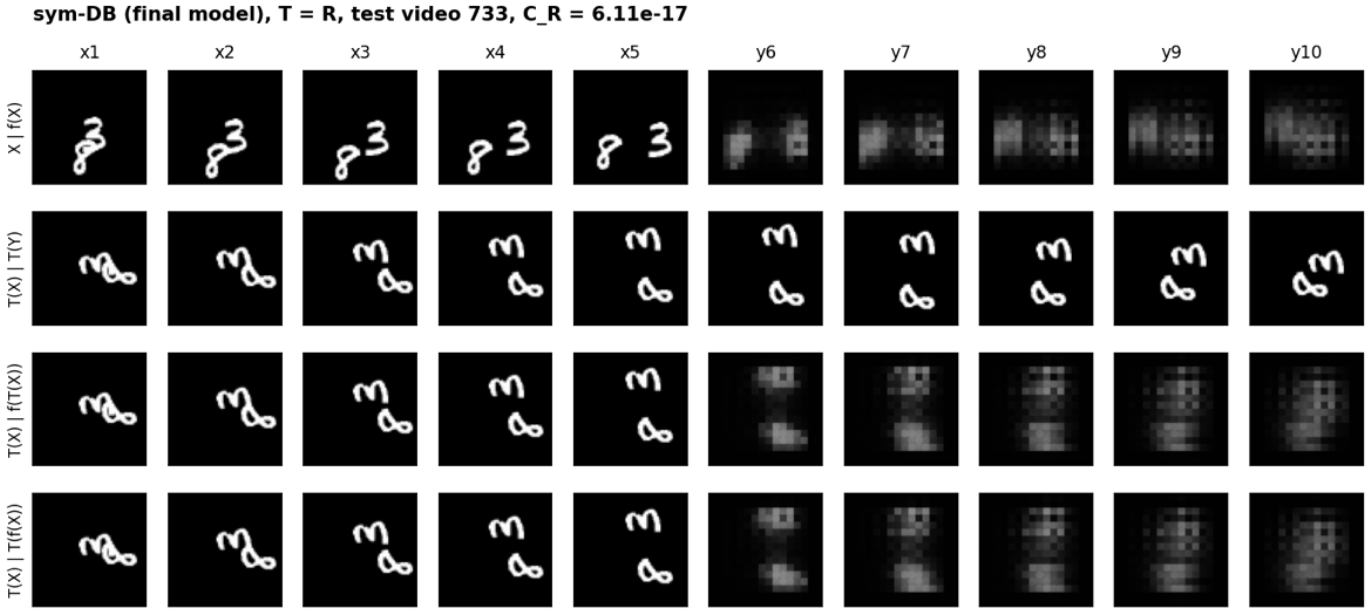
(in the github repository, the files labled final_model are on model sym-D which I ran in case I would not have the time to run D+B. the final model refered to in this report is final_model_DB)

2.15

the observation that catches the eye is sym-B's Cm value of 9.00e-17 compared to model B's Cm of 1.2e-3 and sym-B's Cr value also dropping to near zero while model B's value is at 1.7e-3. Both of these metrics show that built-in symmetry is exact (consistent) even though learning symmetry performs well. Averaging would penalise accuracy across the eight predicitons if the results are not consistent. Learned symmetry does have inconsistency but on a very small scale which is likely to blur them to an extent leading to a lower accuracy which is the case observed by the experiments.

2.16

Figure 6a



As seen in row 3 and 4, the only improvement is in consistency, not the actual digits; the error is higher than previous run too (up to 0.28 from 0.22). addition of the built-in symmetry fixed consistency issue and speed dilation improved on the speed-up issue. However, overall performance is worse since the output is less viable than model A's.

2.17

The final model has indeed learned consistency but has lost on sharpness and digit reproduciblity compared to initial model A. To a skeptic, the model's ability to retain consistency is appealing, however, the lack of legible digits when two separate ones overlap is unappealing and the test the model has not passed.

References and AI

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